

# PHAM DUC DOANH

AI Engineer

## Summary

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Aspiring AI Engineer with a strong interest in Machine Learning, Large Language Models (LLMs), and Generative AI. Experienced in developing AI-powered applications using Python, FastAPI, LangChain, and Retrieval-Augmented Generation (RAG) through academic and internship projects. Passionate about building practical intelligent systems and continuously learning new AI technologies.

## Experience

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**AI Engineer Intern**, MFC Global – Hanoi, Vietnam (Onsite) Apr 2026 – present

- Developed and maintained an AI-powered website for MFC Global, integrating an LLM-based chatbot to provide product information, answer customer inquiries, and assist users throughout the browsing experience.
- Designed a **Retrieval-Augmented Generation (RAG)** pipeline by connecting the chatbot with company documents and knowledge bases, improving the accuracy and relevance of responses.
- Collaborated with the frontend team to integrate chatbot functionality into the website and enhance the overall user experience.
- Participated in deploying and maintaining the application using **Docker**, **Git**, and **MySQL**, while contributing to testing, debugging, and performance improvements.

## Education

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**Vietnam – Hungary Industrial University**, Engineer in Information Technology – Nov 2026

Ha Noi, Viet Nam

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing.

- Current GPA: 3.21/4.00

Achievements:

- Third Prize, University Scientific Research Competition.

## Skills

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- Programming Languages: Python, Java, JavaScript, C#
- AI & Machine Learning: Machine Learning, Deep Learning, Large Language Models (LLMs), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Prompt Engineering
- Frameworks & Libraries: FastAPI, LangChain, Scikit-learn, NumPy, Pandas, Matplotlib.
- Frontend: React, Ant Design, Sass/SCSS
- Databases: MySQL, MongoDB, SQL Server, Redis
- Tool: Git, Docker, Postman
- Cloud & Deployment: VPS
- Backend: REST API Development, Spring Boot, Spring Security, Spring Cloud

## Projects

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### **AI-Powered Labor Law Chatbot**

Developed an AI-powered chatbot that assists users in searching and understanding Vietnamese labor law by combining Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG).

- Developed a web-based chatbot that provides accurate and context-aware answers to labor law questions using a Retrieval-Augmented Generation (RAG) pipeline.
- Built backend APIs with **FastAPI** and integrated Large Language Models (LLMs) to enable natural language question answering.
- Implemented document retrieval from a knowledge base to improve response accuracy and reduce LLM hallucinations.
- Designed and managed a **MySQL** database for storing user information, conversation history, and system data.
- Collaborated on chatbot integration with the web interface and optimized system performance for a smooth user experience.
- **Technologies:** Python, FastAPI, LangChain, OpenAI API, MySQL, RAG.

### **Handwritten Digit Classification using K-Nearest Neighbors (KNN)**

Developed a handwritten digit classification system using the K-Nearest Neighbors (KNN) algorithm to recognize handwritten digits from the MNIST dataset.

- Implemented data preprocessing and feature normalization to improve classification performance.
- Built and evaluated a K-Nearest Neighbors (KNN) model for handwritten digit recognition using the MNIST dataset.
- Experimented with different values of K and distance metrics to optimize model accuracy.
- Evaluated the model using accuracy and confusion matrix to analyze classification performance.
- **Technologies:** Python, Scikit-learn, NumPy, Pandas, Matplotlib.

## Languages

Vietnamese: Native

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**English:** Technical reading proficiency